
Making Sense of Data

Centre, Spread, Shape & the Hunt for Outliers

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Before we can predict or decide anything, we must first *describe* what our data looks like. This companion tells that story in plain language. We learn what kind of data we have, find its “typical” value, measure how spread out it is, read its shape, and finally sketch it as a single tidy picture — the box plot — that even reveals the oddballs hiding in the data. Think of it as learning to read the personality of a dataset before asking it any hard questions.

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1 Statistics: turning raw numbers into decisions

Imagine someone hands you a notebook full of numbers — the minutes you take to get ready each morning, the noise level in an office, the fat content of fast-food sandwiches. On its own, a pile of numbers tells you almost nothing. **Statistics** is the craft of making that pile *talk*.

Statistics is detective work

A detective doesn't just gather clues — they *organise* them, look for patterns, and then draw a conclusion that goes beyond what is directly visible. Statistics does the same with data: we *collect* it, *organise* it neatly, *analyse* it, *infer* something, and finally *decide*. That is why statistics is often called “data-driven decision making.”

What, why, where — for this whole session

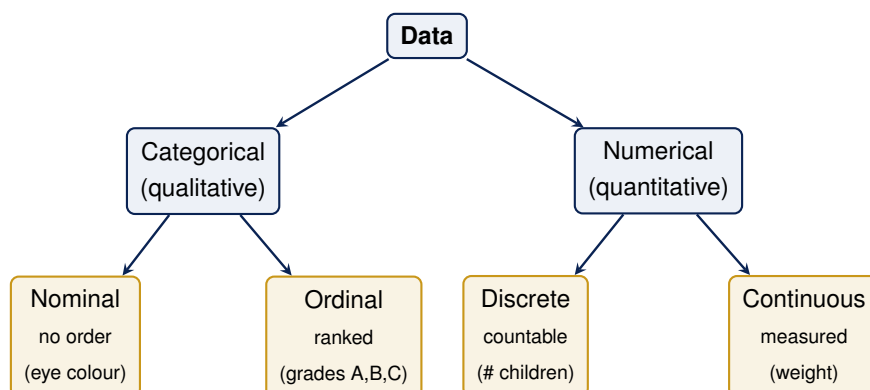
What: *descriptive statistics* — summarising a dataset with a few honest numbers and one good picture. **Why:** a thousand raw numbers overwhelm us; three or four well-chosen summaries let us actually *see* the data. **Where:** every dashboard, every report, every model starts here — you cannot model or predict data you have not first understood.

To understand any dataset we ask four simple questions, and this session answers each in turn: *What kind of data is it? Where is its centre? How spread out is it? What is its shape (and are there any oddballs)?*

2 Step 0 — Know your data before you touch it

What it is. Before computing anything, identify *what kind of numbers* you have. Data splits first into *categorical* (labels) and *numerical* (quantities), and numerical splits again into *discrete* (counted) and *continuous* (measured).

Why it matters. Using the wrong tool on the wrong data type is like trying to average phone numbers — the arithmetic runs, but the answer is meaningless. The data type decides which summaries are even *allowed*.



The four “levels” as a staircase — and what each one lets you do

Numbers carry different amounts of meaning. Picture a staircase from least to most informative; each step *unlocks* more operations:

- **Nominal** — just names, no order. *Eye colour, marital status*. Allowed question: “same or different?” Only the *mode* makes sense.
- **Ordinal** — ordered, but the gaps are unequal. *Grades A, B, C; medals gold, silver, bronze*. You may rank and take a *median*, but “A – B” is not a real amount.
- **Interval** — equal gaps, but *no true zero*. *Temperature in °C*. You may add and subtract; 20° is *not* “twice as hot” as 10°.
- **Ratio** — equal gaps *and* a true zero, so ratios are meaningful. *Weight, age, salary*. Everything is allowed; 40 kg really is twice 20 kg.

The higher the step, the more arithmetic the data will honestly support.

3 Step 1 — Where is the centre? (mean, median, mode)

The question. If you had to describe the whole dataset with a *single* number, what would it be? This “typical value” is a **measure of central tendency**, and there are three classic answers, each with its own personality.

Definition 3.1: Mean, median, mode

For data $y_1, \dots, y_n$:

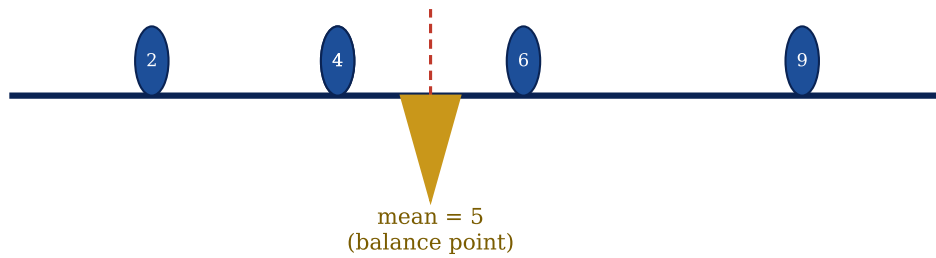
- the **mean** (average) is $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$;
- the **median** is the middle value once the data is sorted (the average of the two middle values when n is even);
- the **mode** is the value that appears most often.

Three ways to find the “middle of the room”

Picture people of different heights standing in a room.

- The **mean** is the *balance point*: if heights were weights on a seesaw, the mean is where it balances. *Every person tugs on it*.
- The **median** is the person in the *middle of the line* when everyone queues up by height — half are shorter, half taller. It only cares about *position*, not size.
- The **mode** is the *most common* height — the one you’d bump into most often.

The mean is where the data would balance on a seesaw



The mean made physical: place each value as an equal weight on a plank. The single point where the plank balances is the mean. Slide one weight far to the right and the balance point must shift to follow it — which is exactly why a lone outlier drags the mean.

Why & where to use each one.

- **Mean** — the workhorse. Every value contributes, it is easy to compute, and it feeds directly into the standard deviation. *Use it* for fairly symmetric data with no wild outliers. *Weakness:* it is easily dragged by extremes (next box).
- **Median** — the robust choice. *Use it* for skewed data or data with outliers (incomes, house prices, waiting times). It is the honest “middle.”
- **Mode** — the only centre that works for *categorical* data. *Use it* when you want the most frequent category (“which product sells most?”).

Worked example 3.1: The three centres on real data — “time to get ready”

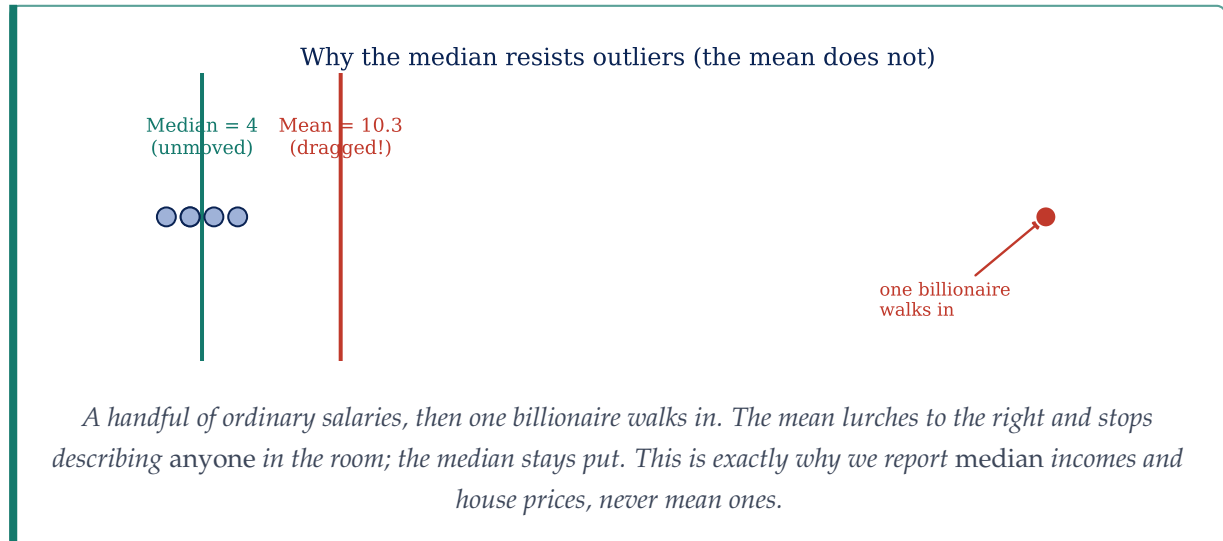
You time your morning routine (minutes) over ten work days: 39, 29, 43, 52, 39, 44, 40, 31, 44, 35. **Mean:** add them ($= 396$) and divide by 10, giving $\bar{y} = 39.6$. **Median:** sort first — 29, 31, 35, 39, **39, 40**, 43, 44, 44, 52 — and average the two middle (5th and 6th) values: $\frac{39+40}{2} = 39.5$. Because the mean (39.6) and median (39.5) almost coincide, the data is nearly symmetric with no extreme outlier — a conclusion we reached *just by comparing two numbers*.

Worked example 3.2: The mode in action — counting server failures

A network manager logs daily server failures over two weeks: 1, 3, 0, 3, 26, 2, 7, 4, 2, 3, 3, 6, 3. The value 3 appears five times — more than any other — so the **mode** is 3. The most *typical* day has three failures. (Notice the mode ignores that lone 26 completely; like the median, it is unbothered by outliers.)

The mean's weakness: it is easily dragged

Because every value pulls on the balance point, a single extreme value (an *outlier*) can drag the mean far from where most of the data sits. The median, caring only about position, barely notices.



A free relationship between the three

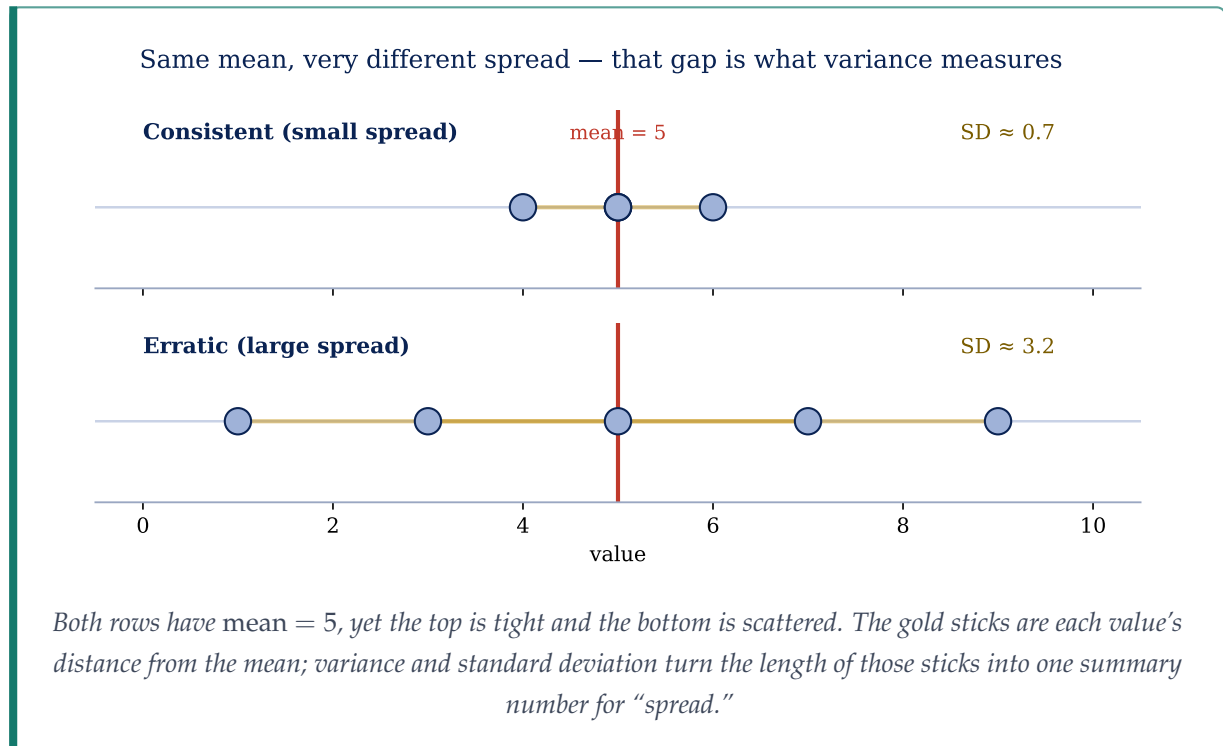
For mildly skewed data the three centres are tied together by a handy empirical rule, $\text{Mode} \approx 3 \text{ Median} - 2 \text{ Mean}$. It is not a theorem — just a pattern observed again and again — but it is a quick sanity check and shows the three centres are different “views” of the same data.

4 Step 2 — How spread out is it? (range, variance, SD)

The question. The centre is only half the story. Two datasets can share the *same* mean yet feel completely different, so we also need to know how *spread out* the values are. This is a **measure of variability**.

Two archers, same average

Two archers both *average* the bullseye. One lands every arrow in a tight cluster; the other sprays them everywhere, with hits far left and far right that happen to average to the centre. Same mean — completely different shooters. **Spread** is what separates the sniper from the sprayer, and it is often what we care about most (consistency, risk, reliability).



The range (the quick-and-dirty measure). Range = max – min. **Why use it:** instant to compute. **Why distrust it:** it listens only to the two most extreme values and ignores everyone in between, so a single outlier fools it completely.

Building the standard deviation, step by step. A far better idea: measure how far each value sits from the mean, then summarise those distances. The distances are the *deviations* $y_i - \bar{y}$. But there is a catch we must design around.

The deviations always cancel to exactly zero

Because the mean is the balance point, the gaps above it exactly cancel the gaps below it: $\sum_i (y_i - \bar{y}) = 0$ *always*. So averaging the raw deviations is useless — the answer is always 0. We need a way to stop the + and – signs from cancelling.

The cure is to **square** each deviation (squares are never negative, so nothing cancels), *average* the squares, and finally take a *square root* to return to the original units.

Key formula 4.1: Variance and standard deviation

The **variance** is the average squared distance from the mean; the **standard deviation** is its square root (back in the original units):

$$\text{Population: } \sigma^2 = \frac{1}{N} \sum_{i=1}^N (y_i - \mu)^2, \quad \text{Sample: } s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2,$$

with $\sigma = \sqrt{\sigma^2}$ (or $s = \sqrt{s^2}$). Read the standard deviation as the *typical distance of a value from the mean*.

Two natural questions, answered

Why square, rather than take absolute values? Squaring is smooth (no sharp corner at zero), which makes the algebra and calculus behind almost all of statistics work cleanly — and it punishes big misses more than small ones. *Why divide a sample by $n - 1$, not n ?* A sample hugs its *own* mean a little too tightly, so dividing by n would slightly *under-estimate* the true spread. Using $n - 1$ (called *Bessel's correction*) gently inflates the estimate to remove that bias — a small “honesty tax” for not having seen the whole population.

Worked example 4.1: Standard deviation, fully worked

Take the nine values 2, 8, 5, 3, 7, 8, 5, 2, 5. **Step 1 — mean:** $\bar{y} = \frac{45}{9} = 5$. **Step 2 — square every deviation from 5 and add them (the “sum of squares”):**

$$\begin{aligned} SS &= (2-5)^2 + (8-5)^2 + (5-5)^2 + (3-5)^2 + (7-5)^2 \\ &\quad + (8-5)^2 + (5-5)^2 + (2-5)^2 + (5-5)^2 \\ &= 9 + 9 + 0 + 4 + 4 + 9 + 0 + 9 + 0 = 44. \end{aligned}$$

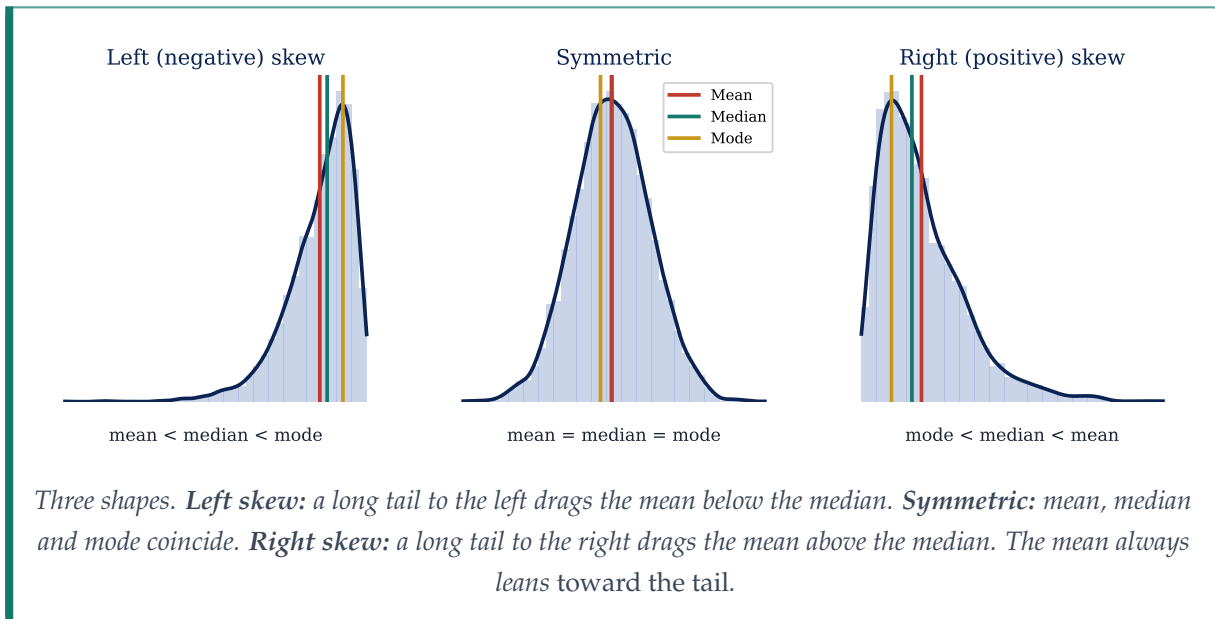
Step 3 — as a sample, divide by $n - 1 = 8$ and take the root:

$$s = \sqrt{\frac{44}{8}} = \sqrt{5.5} \approx 2.35.$$

So a typical value sits about 2.35 away from the mean of 5. Reported together, $(\bar{y}, s) = (5, 2.35)$ already paints a surprisingly complete picture of the data.

5 Step 3 — What is its shape? (skew and symmetry)

The question. Centre and spread still miss one thing: is the data *lopsided*? The shape tells us where the bulk sits and which way the long “tail” stretches — and, crucially, *which centre measure to trust*.



A comet and its tail

A skewed distribution is like a comet: a bright, dense *head* (where most values cluster) and a long, faint *tail* streaming off to one side. The tail's direction *names* the skew (a tail to the right is “right-skewed”), and — like a small moon tugging the tide — that thin tail quietly pulls the mean toward itself.

Read the skew straight off the centre measures

- **Symmetric:** Mean = Median = Mode — e.g. adult heights.
- **Right (positive) skew:** Mean > Median > Mode — e.g. incomes, where a few very high earners stretch the right tail.
- **Left (negative) skew:** Mean < Median < Mode — e.g. scores on an easy test, where a few low scores stretch the left tail.

The mean is the gossip who always leans toward the tail; the median keeps a cool head. **So for heavily skewed data, report the median.** The *gap* between mean and median is itself a quick skewness detector.

6 Step 4 — One tidy picture: the five-number summary & box plot

The big payoff. We can capture a dataset's centre, spread *and* shape with just five well-chosen numbers — and then draw them as a single picture that even reveals the oddballs.

Definition 6.1: Five-number summary, quartiles, IQR

The **five-number summary** is: minimum, Q_1 (the 25% mark), median Q_2 (50%), Q_3 (the 75% mark), and maximum. The **interquartile range**

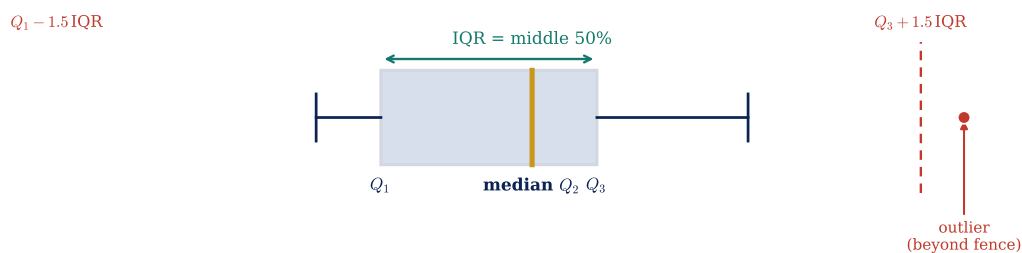
$$\text{IQR} = Q_3 - Q_1$$

is the spread of the *middle* 50% of the data. Because it ignores the extreme quarters at both ends, the IQR is wonderfully resistant to outliers.

Quartiles split the queue into four equal groups

Line everyone up shortest to tallest and cut the queue into four equal-sized groups. The three cut points are Q_1 , Q_2 , Q_3 . The middle two groups — the everyday, unremarkable majority — span the IQR.

Anatomy of a box-and-whisker plot



The box holds the middle 50% (from Q_1 to Q_3); the gold line is the median. The “whiskers” reach to the most extreme values that are still reasonable. Anything beyond the dashed fences at $Q_1 - 1.5 \text{IQR}$ and $Q_3 + 1.5 \text{IQR}$ is flagged as a possible outlier.

Key formula 6.1: The $1.5 \times \text{IQR}$ rule for outliers

A value is a potential **outlier** if it falls outside the fences:

$$\text{Lower fence} = Q_1 - 1.5 \times \text{IQR}, \quad \text{Upper fence} = Q_3 + 1.5 \times \text{IQR}.$$

Where it's used: this exact rule is what statistical software draws as the dots beyond the whiskers in a box plot, and it is a standard first pass for data cleaning before any modelling.

Worked example 6.1: A five-number summary with no outliers

Sorted data: 11, 12, 13, 16, 16, 17, 17, 18, 21 (nine values). The median is the 5th value, $Q_2 = 16$; the quartiles work out to $Q_1 = 12.5$ and $Q_3 = 17.5$, so $IQR = 17.5 - 12.5 = 5$. The fences are

$$Q_1 - 1.5(5) = 5 \quad \text{and} \quad Q_3 + 1.5(5) = 25.$$

Every data point lies between 5 and 25, so this dataset is **free of outliers**, and the five-number summary (11, 12.5, 16, 17.5, 21) sketches the whole story at a glance.

Worked example 6.2: Now with an outlier — watch the rule catch it

Sorted data: 3, 5, 6, 7, 8, 9, 28 (seven values). The median is the 4th value, $Q_2 = 7$; the lower half {3, 5, 6} gives $Q_1 = 5$ and the upper half {8, 9, 28} gives $Q_3 = 9$, so $IQR = 4$. The fences are

$$\text{lower} = 5 - 1.5(4) = -1, \quad \text{upper} = 9 + 1.5(4) = 15.$$

Six values sit comfortably inside $[-1, 15]$, but $28 > 15$ falls past the upper fence — so the rule correctly **flags 28 as an outlier**. This is precisely the dot a software box plot would draw beyond the right whisker.

7 Putting it all together**Try these to lock in the intuition**

- (centre + spread + outliers).** A sample of office noise levels (dBA) ranges from the mid-50s to the low-80s, with most readings clustered in the 60s. *Predict first:* will the mean sit above or below the median? Then compute the mean, standard deviation and IQR, and use the $1.5 \times IQR$ rule to check for outliers.
- (skew detective).** Fat content (grams) of 20 sandwiches: 7, 8, 4, 5, 16, 20, 20, 24, 19, 30, 23, 30, 25, 19, 29, 29, 30, 30, 40, 56. Find the mean, median, Q_1 , Q_3 , the IQR and any outliers. Is the data skewed — and which way? How can you tell from the mean versus the median alone?
- (five-number summary).** Battery life (shots) for digital cameras: 300, 180, 85, 170, 380, 460, 260, 35, 380, 120, 110, 240. List the five-number summary and sketch the box plot.

The thread to the next session

We can now *describe* any dataset: its centre (mean, median, mode), its spread (range, variance, standard deviation, IQR), its shape (skew vs symmetry), and its oddballs (outliers via the box plot). But describing what we *have seen* is only half of statistics. The other half asks: what about the things we *haven't* seen yet — the next customer, the next measurement? Answering that needs the language of *chance*. That is exactly where Session 2 begins: **probability**.